

Experiment 8:

Implement a simple stochastic part-of-speech tagging algorithm using a basic probabilistic model to assign POS tags using python.

Program:

```
import random

noun = ["dog", "cat", "book", "student"]

verb = ["run", "eat", "play", "write"]

adj = ["good", "big", "small", "beautiful"]

sentence = input("Enter words separated by spaces: ")

words = sentence.split()

print("\nPOS Tags:")

for word in words:

    if word in noun:

        print(word, ": NOUN")

    elif word in verb:

        print(word, ": VERB")

    elif word in adj:

        print(word, ": ADJECTIVE")

    else:

        print(word, ":", random.choice(["NOUN", "VERB", "ADJECTIVE"]))
```

Output:

```
for word in words:
    if word in noun:
        print(word, ": NOUN")
    elif word in verb:
        print(word, ": VERB")
    elif word in adj:
        print(word, ": ADJECTIVE")
    else:
        print(word, ":", random.choice(["NOUN", "VERB", "ADJECTIVE"]))
```

.. Enter words separated by spaces: dog play apple beautiful

POS Tags:
dog : NOUN
play : VERB
apple : NOUN
beautiful : ADJECTIVE

